Counts, normalization & differential expression

Pseudobulk	Sum all of one donor's hepatocyte UMIs into a single profile, so the donor — not the cell — is the statistical unit (cells in a donor are not independent).
CPM (counts per million)	Each gene as a share of the donor's total counts; simple but compositional — if a few genes surge, everyone else's share mechanically shrinks.
TMM (trimmed mean of M-values)	Pick one sample as reference, take each other sample's gene-wise log-ratios to it, trim the extreme movers, average the rest → the library's scaling factor, so surging genes can't distort it (composition-robust).
edgeR	The R / Bioconductor package that runs our per-gene differential-expression test on the pseudobulk counts.
GLM (generalized linear model)	A regression linking a predictor (here: fibrosis stage) to a response that isn't normally distributed — for counts, through a negative-binomial noise model.
NB quasi-likelihood GLM	edgeR's engine: model each gene's counts as negative-binomial (mean + overdispersion) and fit a quasi-likelihood GLM that estimates the extra variability robustly → calibrated p-values with few replicates.
filterByExpr	edgeR's pre-filter: keep a gene only if it has enough reads in enough samples of the smallest stage group, else drop it (≈21,000 of ~36k kept). Low-count genes carry no statistical power and only inflate the multiple-testing burden, so removing them sharpens the real hits.
Binomial down-sampling	Randomly thin every nucleus to a common 1,500 UMIs (each molecule kept with equal probability) so sequencing depth can't drive detection.

Gene-set tests & statistics

GSEA (pre-rank)	Rank every gene by its fibrosis-trend statistic, then test whether a gene set clusters near one end of that ranking instead of being scattered.
CAMERA	A competitive set test: is this set shifted more than the rest of the genome? — and it corrects for correlation between the set's genes.
ROAST / mroast	A self-contained set test: does the set move away from zero on its own (p-value from rotations)? — ignores the rest of the genome, more conservative.
competitive vs self-contained	Competitive asks “more shifted than other genes?”; self-contained asks “different from no change at all?” — a set can pass one and fail the other.
TOST (two one-sided tests)	An equivalence test: not “is there a difference?” but “is the difference small enough to rule out anything bigger than a set bound?” — lets us say “no large change,” not just “nothing significant.”
Cramer's V	A 0–1 score of association between two categorical variables (e.g. sequencing run vs tissue source): 0 = independent, 1 = perfectly confounded.
Spearman correlation	Correlation of ranks — captures any monotonic trend, robust to outliers and non-linear scaling (used for donor-level fibrosis trends).
Partial correlation	Correlation between two variables after removing a third's effect (e.g. detox vs fibrosis, controlling for age).

Tools & data terms

decontX	Models each droplet as native RNA + a shared ambient “soup,” estimates each cell's ambient fraction, and subtracts it → a decontaminated count matrix.
SCT (SCTransform) — why we dropped it	Seurat normalization that swaps raw UMIs for model-smoothed values. Our two core questions are defined on integer molecules, which SCT destroys: (a) marker co-detection asks “does this nucleus carry ≥ 2 real molecules of marker X?” — a “≥ 2 UMI” call is meaningless on smoothed residuals; (b) ambient sensitivity must tell 0 vs 1 vs 2 true molecules apart, but SCT pools / imputes across cells and would manufacture apparent co-expression. So we work on raw counts.
Ambient RNA (“soup”)	Cell-free RNA from lysed cells floating in the droplet suspension; every droplet absorbs some, so genes from abundant cell types leak into other cells.
Ductular reaction	The expansion of bile-duct / cholangiocyte cells in cirrhosis — it makes the soup richer in biliary RNA exactly at F4.
Doublets	Two nuclei captured in one droplet and read as a single “cell” — a hepatocyte + cholangiocyte doublet fakes co-expression.
BCV / dispersion	Biological coefficient of variation — how much a gene's counts vary beyond Poisson noise; edgeR estimates it to avoid false positives.
Anchor classification	A yes/no detection call: a nucleus is “PC” if it detects GLUL/CYP3A4 at ≥ 2 UMI — robust to a level drop, which is why the fraction can stay flat while the program dims.

The confounder battery, explained (1 / 2)

Tissue source	Could fool us because disease stage is tangled with how the tissue was obtained — healthy & end-stage are deceased-donor organ cubes / explants while F0–F4 are needle biopsies; since the ends aren't acquisition-matched, the verdict is to exclude them and analyze biopsy-only F1–F4.
Procurement stress	Tissue handling switches on stress genes that could masquerade as disease; we measured immediate-early + heat-shock genes across 6 cell lineages — hepatocytes spike ~18× but endothelium (which has no zonation program) spikes the same ~18×, so the verdict is organ-wide handling stress, not a zonation change.
Sequencing batch	If sequencing runs line up with stage, batch effects could mimic disease; the run↔source association is high (Cramér's V = 0.84) but run↔fibrosis within biopsies is only 0.40 — so batch is fully confounded at the ends, yet the biopsy axis stays estimable.
Lobe (R / C / L)	Explants sample several liver lobes (right / caudate / left) and lobe differences could skew detection; comparing marker detection across the three lobes shows it is lobe-invariant, so lobe is not the driver.
Sequencing depth	Deeper-sequenced nuclei detect more genes, so depth alone can fake “more zonation”; we binomially down-thinned every nucleus to a common budget (1,000 / 1,500 / 3,000 UMIs) and the PC-share stayed flat at every budget (SD only 0.006–0.010), so depth is not driving it.

The confounder battery, explained (2 / 2)

Ambient RNA (“soup”)	Ambient RNA from abundant genes (e.g. albumin / ALB) could create fake co-expression; across 38 donors the correlation between a donor's ALB ambient share and their dual (co-expression) fraction is +0.04 — essentially zero — so ambient does not drive co-expression.
Cholangiocyte mis-annotation	If some “hepatocytes” are actually mislabeled cholangiocytes they would fake a biliary signal; within every anchor class the cholangiocyte markers are near-absent ($KRT19^+ \le 0.1\%$, $EPCAM^+ \le 0.3\%$), so the anchor cells are not contaminating cholangiocytes.
Ploidy / complexity	Cells of different complexity (number of genes detected, nFeatures) could track stage and bias detection; mean nFeatures by stage is non-monotone (2218 / 3063 / 3169 / 2641 / 1984), so complexity does not track stage.
Depth-match discard	Down-thinning discards nuclei that fall below the UMI budget, and if those discards were pericentral-biased it would skew the result; the 3.5–7.8% dropped are uniformly lower-detection cells — depth-driven, not PC-biased.
Clinical covariates	Demographics like age correlate with fibrosis and could explain the detox trend; the partial correlation of detox with fibrosis controlling for age is −0.32 (age–fibrosis = +0.43), so demographics do not explain the trend.

Distinctions we kept confusing

Dimming vs de-zonation	De-zonation = cells change identity (refuted); dimming = cells keep identity but the program's level drops (the one real change). Radio: genre change vs volume down.
Anchor genes vs detox genes	Identity is a detection call on GLUL + CYP3A4; the detox dimming is measured on separate genes (CYP2E1, CYP1A2 …), so identity stays flat while the module's level falls.
Depletion vs turn-off	Depletion = an anchor (pole) box shrinks; turn-off = the null (double-negative) box grows — different boxes, opposite directions, neither implies the other.
Compression vs dimming	Compression = the gradient's shape flattens (cells drift to the middle); dimming = the level drops with the shape kept.
Relative vs absolute metric	A z-score / correlation reads the gradient's shape (scale-free); only counting molecules reads how many cells sit where.
Donor as the replicate	The ~47 donors are the independent units; cells within a donor are pseudoreplicates → summarize per donor, or you fake significance (SE ≈ SD/√n).
Biliary = compositional confounder	At F4 a ductular reaction adds cholangiocytes whose ambient RNA enriches hepatocyte droplets — so a “hepatocyte biliary” signal can rise from tissue composition, not transdifferentiation. decontX removes most hits; EPCAM/SPINT2/B3GNT3 survive → a rare real state not excluded.
Equivalence bound vs proof	TOST excludes shifts > ±19 pp; a ≤ 10-pp drift stays possible — “no large change,” not “exactly zero.”

Donor figures: display & honest claims

What the bar shows	Dots = every donor (the spread + outliers — nothing hidden). The error bar shows the precision of the center — NOT the spread again (the dots already show that).
SD / IQR / whiskers	Spread of donors (descriptive); redundant with the dots — and median + IQR is literally the box plot's box.
SEM = SD/√n	Precision of the mean; tight because of the ÷√n. ← what we use: a big donor spread still gives a well-pinned mean.
95% CI = SEM × t	Also precision, but t ≈ 3.2 at n = 4 blows it up → reads “uncertain” and fights the “confident” message.
Mean, not median	The inference (TOST, SE ≈ 4 pp) is mean-based; a median center would describe a different quantity than the test. Median + IQR is the box again.
Honest, not stat-picking	Raw dots shown; equivalence stated both ways; and a real change WAS found (detox dimming, ρ = −0.48) → a genuine null, not a power failure. Soft spot: F4 n = 4.
✗ Don't claim	“zonation is intact / preserved” — can't prove a negative; underpowered (F4 n = 4).
✓ Do claim	No detectable de-zonation signal; a large shift (> ±19 pp) is excluded (TOST); a subtle ≤ 10 pp drift is not (F4 n = 4).